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KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

Department of information and communication technology

FINAL EXAM

Course Title: CYBERCRIME INVESTIGATION AND FORENSICS 2

Course Code: CYS 340

Instructor Name: MR NGOSA GATHAM

Moderator Name: .....

Date: .....

Duration: 3HRs

Total Marks: 60

General Instructions to Candidates:

1. Answer question 1 and any other three
2. No unauthorized materials are allowed
3. No electronic devices allowed unless specified.
4. Start each answer on a new page.
5. Write clearly and legibly.

Exam Instructions:

Any student that will be caught with foreign material, copying or behaving in a manner viewed as examination malpractice/misconduct, will be subjected to disciplinary hearing or disqualification.

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Question 1

A malware sample injects itself into system processes and hides from disk scans.

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(2) a) As a forensic analyst, Explain step-by-step the process of analyzing volatile memory in this case using memory forensics. (3 marks)

(2) b) Discuss four unique conditions where only memory forensics applies (4 marks)

(4) c) Describe at least five main formats of memory data essential for forensic investigations. (5 marks)

(2) d) Discuss three methods of memory data acquisition that you may opt to use in this case (3 marks)

Question 2

During a cyber-attack investigation, you are tasked with analyzing a seized hard drive that contains deleted files suspected to be related to illegal activities

14/15

(3) a) Explain the possible types of deletion the suspect could have used. (3 marks)

(6) b) Outline the typical steps involved in file carving. (6 marks)

(5) c) Discuss challenges faced during file reconstruction like this one and propose solutions to overcome such. (6 marks)

Question 3

Investigators are asked to analyze a cyber-attack in a hybrid cloud model.

a) What challenges are they likely to face, and what measures should they adopt (6 marks)

b) With the aid of a diagram, discuss cloud service models from a forensic perspective (3marks)

c) Compare and contrast cloud forensics to traditional digital forensics(6 marks)

Question 4

1. A hacker used a rootkit to maintain persistence in a victim's machine

12/15

(2) a) As a forensic analyst, Describe how malware forensics can uncover such hidden threats?(5 marks)

(5) b) Identify five types of malware common in forensics, describe their characteristics and how they can be detected. (5 marks)

(3) c) Describe three proactive measures to protect systems from malware attacks (3 marks)

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(2) d) With clear examples, distinguish signature-based from heuristic-based malware detection techniques. (2 marks) ✓

Question 5

You are asked to present forensic evidence in court.

a) Explain how adherence to digital forensic regulations ensures admissibility. (4 marks)

b) Explain Locard's Exchange Principle and its significance in digital forensics(2 marks),

c) Discuss at least five organizations that govern digital forensics internationally (5 marks). WGDF NIST AAS LAB

d) If a fraud case involving online banking has been reported. As a cyber-forensic investigator (CFI), describe your roles in solving this case. (4 marks)

Question 6

An investigator finds multiple multimedia files that have been tampered with in an online fraud case

(2) a) Explain how multimedia forensics helps in establishing the integrity of evidence (5 marks) ✓

8/15 (3) b) Describe methods used to analyze images, audio, and video files (3 marks) ✓

(1) c) Discuss why understanding the five elements of multimedia aids in collecting reliable digital evidence. (5marks) ✓

(2) d) Identify and describe two common cybercrime investigation tools used multimedia forensics (2 marks) ✓

Q1 = 10
Q2 = 14
Q4 = 12
Q6 = 8
44

44/60

EXAM = 44
CA = 36
TOTAL = 80
GRADE = A

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